

AquaSense — Automatic Touchless Water Dispenser

Phase 1 Project Report

Electronic Design, Embedded Programming & 3D Enclosure Development

Department of Electronics and Communication Engineering
REVA University

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Team: Kushwanth & Likith M.
Mentor: Mr. Mahesh Devagiri

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1. Project Title

AquaSense — Automatic Touchless Water Dispenser

AquaSense is the project title for the Phase 1 development of an Arduino-based touchless water-dispensing system.

2. Project Identity

Item	Details
Phase	Phase 1 — Electronic Design, Embedded Programming & 3D Enclosure Development
Department	Electronics and Communication Engineering
Institution	REVA University
Project Developer	Kushwanth P
Team	Kushwanth & Likith M.
Mentor	Mr. Mahesh Devagiri

3. Project Description

AquaSense is an Arduino-based touchless water dispenser designed to operate a water tap without direct physical contact. An HC-SR04 ultrasonic sensor detects a user's hand, while an Arduino Uno processes the detection and controls two servo motors that operate the dispensing mechanism. A 16×2 I2C LCD provides system feedback and distance information. Phase 1 focuses on the electronic design, embedded software, system integration planning, and development of a custom 3D-printable enclosure.

4. Core Technologies

Technology	Purpose
HC-SR04 ultrasonic sensor	Detects the user's hand and measures distance.
Arduino Uno	Processes sensor input and controls the system.
16×2 I2C LCD	Displays status and distance information.
Dual servo motors	Operate the tap-dispensing mechanism.
Custom enclosure	Provides a single-box mechanical housing for the system.

System Flow

- 1. Detection:** HC-SR04 ultrasonic sensor detects the user's hand.
- 2. Processing:** Arduino Uno evaluates the measured distance.
- 3. User Feedback:** 16×2 I2C LCD displays distance and system status.
- 4. Actuation:** Dual servo motors operate the tap mechanism when the detection condition is satisfied.

5. Key Phase 1 Features

- Touchless hand detection.
- 15 cm detection threshold.
- Arduino Uno control.
- Dual-servo actuation.
- 16×2 I2C LCD feedback.
- Approximately 3-second dispensing cycle.

- Custom single-box enclosure.
- Mechanical servo-linkage concept.

The 15 cm detection threshold and approximately 3-second dispensing sequence are design parameters documented for the Phase 1 system. They should be confirmed through physical measurement during the subsequent testing stage before being presented as experimentally validated performance results.

6. Phase 1 Scope

Phase 1 concentrates on electronic design, Arduino programming, system architecture, mechanical integration planning, and 3D enclosure development. Physical validation of the final dispensing mechanism remains part of the subsequent testing stage.

This scope intentionally separates design and documentation work from measured prototype performance. Quantitative claims such as detection accuracy, dispensing duration, servo response, reliability, and power behavior should be finalized only after the physical prototype has been assembled and tested with recorded measurements.

7. Project Development Journey

The project followed a staged development path from identifying the need for touchless operation to preparing the system for future physical validation. The journey reflects the progression from problem definition and concept generation through electronic design, software development, mechanical integration, CAD refinement, and testing planning.

Stage	Development activity
Problem	Identify the need for hygienic, contactless water dispensing.
Concept	Define an Arduino-based system using hand-distance detection and servo actuation.
Circuit & Hardware	Plan the HC-SR04, Arduino Uno, LCD, power, and dual-servo connections.
Arduino Software	Develop the logic for detection, display feedback, and dispensing control.
System Architecture	Integrate sensing, processing, user feedback, and actuation into one workflow.
Mechanical Design	Plan the tap interface, servo arrangement, linkage, and enclosure requirements.
Single-Box CAD	Develop the custom 3D-printable box and lid as a unified enclosure.
Servo Linkage	Design the mechanical connection that transfers servo motion to the tap mechanism.
Final Render	Prepare the final CAD presentation and visual documentation.
Testing	Prepare test cases and planned/simulated testing documentation; physical validation follows in the next stage.
Future Phase 2	Fabricate, assemble, measure, calibrate, and validate the complete dispensing prototype.

Problem → Concept → Design → Development → Testing → Product → Future

Conclusion

AquaSense Phase 1 establishes a complete design foundation for a touchless water dispenser. The project combines ultrasonic sensing, Arduino control, LCD feedback, dual-servo actuation, and a custom single-box enclosure. The documented design is ready to proceed toward fabrication and physical validation in Phase 2, where the dispensing mechanism and its quantitative performance will be measured and improved.